Graph Based Classification of Social Media Discourse into Psychological Impact Categories Under Reduced Supervision

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***Abstract*—Social media generates vast quantities of daily interactions that influence user emotions, cognition, and decision making. Understanding the psychological impact of such interactions demands modelling not just the entities involved, but the relationships between them. This study proposes a graph-based framework to evaluate the psychological impact of social media interactions on users. Discourses were extracted from YouTube and Hacker News using their official APIs and structured as a directed graph, with posts, comments, and replies as nodes connected by two categories of edges: conversational reply edges and semantic similarity edges. Node-level multi-class classification was performed using Graph Convolutional Networks (GCN), GraphSAGE, and Graph Attention Networks (GAT) to assign each node one of three psychological impact labels: *Good*, *Bad*, or *Nuanced*. The results show that graph-based models consistently outperform traditional text-only approaches by leveraging relational context in social media discourse. This study draws a deliberate distinction between the psychological impact content leaves on the reader and the sentiment it expresses.**

***Index Terms*—Discourse, Psychological Impact, Node-level Multi-class Classification, Semantic Similarity, Graph Neural Networks, Reduced-Supervision Learning**

I. Introduction

The human brain releases dopamine that motivates repeating activities associated with happiness and relaxation—a survival mechanism that originally helped mankind seek food and safety. In recent years this mechanism is triggered easily by activities as passive as scrolling through social media, reinforcing continued platform engagement independent of the quality of content consumed.

Social media platforms leverage this biological reward mechanism to maximise user engagement, often prioritising polarising content that keeps users active for longer periods. The psychological trace left by such content is not captured by sentiment polarity alone. A reply that is positive in sentiment can still induce anxiety, confirmation bias, or desensitisation in the reader—effects that sentiment labels never surface.

Most existing systems conflate sentiment with impact. This study argues that the two are distinct: sentiment is a property of the text relative to the reader's prior expectations, whereas psychological impact is the effect the content leaves on the reader regardless of those expectations. Classifying content by impact therefore requires a richer contextual signal than word-level polarity provides.

The paper formulates this as a node-level multi-class classification problem on a conversational graph, where the graph structure itself carries relational evidence that a standalone text classifier cannot access. Data were gathered from YouTube and Hacker News across seven domains. Initially a four-way labelling strategy was planned (Good Impact, Bad Impact, Mature, Toxic). Upon observing empirical overlap between *Good* and *Mature*, and between *Bad* and *Toxic*, the labelling scheme was consolidated to a three-class scheme (*Good*, *Bad*, *Nuanced*) that reflects psychologically meaningful boundaries without redundancy.

Node-level classification was carried out using GCN [1], GraphSAGE [2], and GAT [3]. Models were trained under a reduced-supervision setting in which only 20% of labelled nodes contributed to loss computation, simulating the annotation-budget constraints common in real deployment scenarios. The remainder of this paper is organised as follows: Section II reviews related work; Section III describes the proposed methodology; Section IV presents and discusses results; Section V concludes.

II. Related Work

A. Graph Neural Networks

Kipf and Welling [1] proposed the Graph Convolutional Network (GCN), which operates on graphs via spectral convolutions, learning features that jointly encode graph structure and node attributes. GCN requires the full graph to be present at training time and does not generalise to unseen nodes. Hamilton et al. [2] addressed this through GraphSAGE, an inductive framework that generates embeddings for new nodes by sampling and aggregating features from their local neighbourhood—a property directly relevant to the study's setting, where the conversational graph is dynamic and thread boundaries vary. Veličković et al. [3] introduced Graph Attention Networks, which compute attention weights over existing

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edges to assign different importance to different neighbours implicitly, without requiring the graph structure to be fixed a priori.

Recent advances in Graph Neural Networks (GNNs) have been systematically studied, highlighting key architectural paradigms such as spectral and spatial methods, neighborhood aggregation mechanisms, and their effectiveness across relational learning tasks. These insights establish GNNs as a robust framework for modeling structured dependencies, motivating their use in capturing interaction and semantic relationships in social media discourse graphs [4].

B. Sentiment Analysis and Semi-Supervised Learning

Mao et al. [5] surveyed sentiment analysis at document, sentence, and aspect levels, covering techniques and use cases across healthcare, business, and government. Their survey did not address practical strategies for partially labelled corpora. Lee et al. [6] demonstrated that deep neural networks trained on small labelled subsets of sentiment data can perform competitively when unlabelled data is handled carefully, noting that careless inclusion of unlabelled samples degrades performance rather than improving it.

C. Graph-based Approaches to Misinformation and Fake News

Kim and Ko [7] detected fake news at document level by extracting sentence features with a Bi-LSTM, building a sentence-level graph using attention, and ranking and classifying the result. Benamira et al. [8] extended graph-based fake news detection to a semi-supervised setting, training GCN and attention-based GNN models with limited annotations on a graph whose nodes are GloVe embeddings and edges are established by embedding proximity.

D. GNNs in Social Network Analytics

Khemani et al. [9] proposed an Improved GCN for emotion analysis in social media text, constructing the graph using Pointwise Mutual Information to capture word-level co-occurrence as a proxy for semantic relatedness.

E. Gap Addressed

Existing work concentrates on sentiment polarity, emotional tone, and veracity classification. The psychological impact of content—which can diverge sharply from both sentiment and toxicity—has received little systematic attention. This study addresses that gap directly.

III. Methodology

A. Data Collection

Data were collected from Hacker News [10] and YouTube [11] using their official APIs across seven domains: technology, business, climate change, mental health, politics, healthcare policy, and social justice. Domain diversity was intentional—impact labels are not uniformly distributed across domains, and a domain-skewed corpus would bias the classifier toward the content patterns of whichever domain dominates. The collection is summarised in Table I.

TABLE I
DATASET DESCRIPTION

Data Source	Posts	Comments & Replies
YouTube	227	17,269
Hacker News	84	6,573
Total	**311**	**23,842**

The raw collection yields 24,153 content nodes (Table I). The final graph contains 23,873 nodes; the additional 280 nodes were dropped due to either missing content or non-informative text. Given their negligible proportion relative to the full dataset, their removal does not materially affect training performance.

B. Data Preparation

1) Unified Schema: Each platform returned data in its own schema. Algorithm 1 describes how both sources were merged into a unified node table U, whose schema is detailed in Table II.

TABLE II
SCHEMA OF UNIFIED NODE TABLE U

Column	Type	Description
NodeID	String	Platform + NodeType + RowIndex
Platform	String	`HN` or `YT`
NodeType	String	`POST` or `CMT`
Text	String	Raw text content
Category	String	Domain; `UndefinedCategory` if absent
ParentID	String	Parent NodeID; `ROOT` for posts
ThreadID	String	NodeID of the thread-root post

2) Annotation: Labels were assigned using BERT [12] as a zero-shot classifier, prompted with explicit definitions of the three categories. While efficient, this introduces noise due to model bias and the subjective nature of psychological impact classification. The dataset is therefore treated as weakly supervised, where labels provide approximate supervision. Confidence scores associated with predictions are used as a heuristic indicator of label reliability, offering a practical trade-off between scalability and annotation quality.

A stratified sample of the dataset was manually reviewed to assess label quality. Agreement between model-generated labels and human annotations was approximately 90%, indicating reasonable alignment despite the subjective nature of the task.

The three label definitions are:

- **Good** — Text that does not induce psychological discomfort or distress and reflects a benign or constructive outlook.
- **Bad** — Text that carries a negative or harmful outlook, induces discomfort, or promotes polarising or misleading perspectives.
- **Nuanced** — Text that requires critical evaluation and cannot be readily categorised as either Good or Bad, often containing mixed or context-dependent signals.

Algorithm 1 Unified Node Table Construction

Require: HNP, HNC, YTP, YTC {Posts and comments from each platform}
Ensure: Unified node table U

$U \leftarrow \emptyset$
for $D \in \{HNP,\ HNC,\ YTP,\ YTC\}$ **do**
  $P \leftarrow$ ExtractPlatform(D) {`HN` or `YT`}
  $T \leftarrow$ ExtractType(D) {`POST` or `CMT`}
  for each record $r \in D$ **do**
    Initialise node u
    u.NodeID $\leftarrow$ Concat($P,\ T,\ r$.RowIndex)
    u.Platform $\leftarrow P$; u.NodeType $\leftarrow T$
    if $T =$ `POST` **then**
      u.ParentID $\leftarrow$ `ROOT`
    else
      u.ParentID $\leftarrow$ GetParentID(r.ParentRef) {Resolves platform parent ref to a NodeID in U}
    end if
    if r.Category = NULL **then**
      u.Category $\leftarrow$ `UndefinedCategory`
    else
      u.Category $\leftarrow r$.Category
    end if
    u.ThreadID $\leftarrow$ AssignThread(r)
    $U \leftarrow U \cup \{u\}$
  end for
end for
return U

The boundary between *Good* and *Nuanced* is intentionally conservative: when disagreement arises between model-generated labels and manual review, the assignment defaults to *Nuanced* to preserve potentially ambiguous content that may carry subtle or context-dependent impact.

3) Preprocessing: Text was cleaned by removing URLs, HTML tags, emojis, and stop-words. Emojis and URLs were removed not as a generic step but because pilot experiments showed they inflated embedding distances between otherwise semantically similar nodes, producing spurious semantic edges during graph construction. Text was normalised before embedding generation.

C. Graph Construction

1) Graph Definition: The discourse is represented as a directed graph $G = (N, E)$ where N is the set of content nodes and E is the union of two disjoint edge sets, as given in (1):

$$E \ = \ E_1 \ \cup \ E_2 \tag{1}$$

where E_1 is the set of reply-based edges and E_2 the set of semantic edges, each described below.

2) Reply-based Edges (E_1): Let GetParentID(n_i) return the ParentID of node n_i and GetNodeID(n_j) return the NodeID of n_j. Reply edges are directed from child to parent as defined in (2), preserving the natural information flow in a conversational thread:

$$E_1 = \big\{(n_i \to n_j) \ \mid \ \text{GetParentID}(n_i) = \text{GetNodeID}(n_j)\big\} \tag{2}$$

The child-to-parent direction was chosen deliberately: context flows upward toward the root post, so this direction ensures each node aggregates information from the content it is responding to during message passing.

3) Semantic Edges (E_2): A graph built on E_1 alone is sparse. Each post initiates a separate thread and inter-thread connectivity is zero by construction. Semantic edges address this by connecting nodes whose text embeddings are close in cosine similarity space, introducing long-range dependencies across sub-graphs as given in (3):

$$E_2 = \big\{(n_i \to n_j) \ \mid \ \tfrac{h_i \cdot h_j}{\|h_i\| \, \|h_j\|} \geq \tau, \ i \neq j\big\} \tag{3}$$

where $h_i, h_j \in \mathbb{R}^{768}$ are the pre-computed sentence embeddings of nodes n_i and n_j respectively, and τ is the similarity threshold. Algorithm 2 details the construction procedure.

Algorithm 2 Semantic Edge Construction

Require: Node set N; threshold τ; pre-computed embeddings $\{h_i\}$
Ensure: Semantic edge set E_2

$E_2 \leftarrow \emptyset$
for each ordered pair (n_i, n_j) where $i \neq j$ **do**
  $S_{ij} \leftarrow \dfrac{h_i \cdot h_j}{\|h_i\| \, \|h_j\|}$
  if $S_{ij} \geq \tau$ **then**
    $E_2 \leftarrow E_2 \ \cup \ \{(n_i \to n_j)\}$
  end if
end for
return E_2

a) Choice of Similarity Threshold.: The threshold $\tau = 0.75$ was selected to balance graph connectivity and semantic precision. A sensitivity check conducted on a 1,000-node subset showed that decreasing τ from 0.95 to 0.75 increased the number of semantic edges from 3 to 1,209, indicating a rapid increase in connectivity at lower thresholds, as summarised in Table III. Higher values of τ produce semantically stricter but sparser graphs that limit effective message passing; lower values introduce semantically weaker edges that increase noise. Empirically, $\tau = 0.75$ provides a reasonable trade-off between meaningful semantic relationships and sufficient information flow. Due to the $\mathcal{O}(|N|^2)$ complexity of pairwise similarity computation, exhaustive tuning across all threshold values was not performed on the full graph.

b) Computational Complexity.: Algorithm 2 is $\mathcal{O}(|N|^2)$. For $|N| = 23{,}873$ this amounts to approximately 5.7×10^8 pairwise comparisons. Embeddings were pre-computed and cached [13], [14] before the edge construction loop to avoid redundant forward passes.

TABLE III
EFFECT OF SIMILARITY THRESHOLD τ ON SEMANTIC EDGE COUNT (1,000-NODE SUBSET)

τ	**Number of Semantic Edges**
0.95	3
0.90	13
0.85	44
0.80	267
0.75	1,209

4) Global Virtual Node: Semantic edges reduced sparsity considerably but the graph diameter remained high. A global virtual node N_0 was introduced as defined in (4), connecting to every node $n_i \in N$ via a directed edge:

$$(N_0 \rightarrow n_i), \quad \forall\, n_i \in N \tag{4}$$

This reduces the effective diameter to at most two hops from any node, enabling each node to access graph-wide context during message passing. A trade-off exists: N_0 injects the same global signal into all nodes simultaneously, which may partially suppress local neighbourhood signals. The net effect on classification accuracy was positive in the experiments conducted, suggesting the benefit of global context outweighed this suppression at the graph scale considered for the study.

5) Final Graph Properties: The graph is directed, contains no self-loops, and is heterogeneous (two edge types: E_1 and E_2, as defined in (1). Text embeddings were generated using the Sentence-BERT framework [13] with the distiluse-base-multilingual-cased-v2 sentence transformer [14], accessed through the Hugging Face Transformers library [15]. This model was chosen for its multilingual capability, accommodating the code-switched and informal language frequent in social media text from both platforms. Embeddings were normalised to zero mean and unit variance before training. The composition of the final graph is given in Table IV.

TABLE IV
COMPOSITION OF THE CONSTRUCTED GRAPH ($\tau = 0.75$)

Property	**Value**
Number of nodes	23,873
Reply-based edges (E_1)	23,833
Semantic edges (E_2)	1,209

Table IV summarises the graph structure used in all subsequent experiments. The semantic edges, while far fewer than reply-based edges, play a critical role in reducing inter-thread isolation.

D. Graph Neural Network Formulations

Let $G = (N, E)$ be the constructed graph with $|N| = M$ nodes. Each node n_i has an initial feature vector $h_i^{(0)} \in \mathbb{R}^d$, where $d = 768$ is the embedding dimension. Let $\mathcal{N}(i)$ denote the set of neighbours of node n_i. The forward pass of each GNN layer produces updated representations $h_i^{(l)}$ at layer l, which are ultimately passed through a softmax classifier to produce a probability distribution over the three impact classes $\mathcal{C} = \{\text{Good}, \text{Bad}, \text{Nuanced}\}$.

1) Graph Convolutional Network (GCN): In GCN [1], the layer-wise propagation rule aggregates the normalised sum of neighbour features. At layer l, the hidden representation of node n_i is updated as given in (5):

$$h_i^{(l)} = \sigma\left(W^{(l)} \sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\tilde{d}_i\, \tilde{d}_j}}\, h_j^{(l-1)}\right) \tag{5}$$

where $\tilde{d}_i = 1 + |\mathcal{N}(i)|$ is the degree of node n_i with a self-loop added, $W^{(l)} \in \mathbb{R}^{d_l \times d_{l-1}}$ is a trainable weight matrix at layer l, and $\sigma(\cdot)$ is a non-linear activation function (ReLU in the implementation of methodology described in Section III). The symmetric normalisation term $(\tilde{d}_i\, \tilde{d}_j)^{-1/2}$ prevents high-degree nodes from dominating aggregation. Three such layers were stacked, with hidden dimension 128 at each intermediate layer, producing a final node representation $h_i^{(3)} \in \mathbb{R}^3$ fed to a softmax classifier.

2) GraphSAGE: GraphSAGE [2] replaces full-neighbourhood aggregation with a sampled, fixed-size neighbourhood to support inductive learning. At layer l, the representation is updated as shown in (6):

$$h_i^{(l)} = \sigma\Big(W^{(l)} \cdot \text{CONCAT}\Big(h_i^{(l-1)},\ \text{AGG}\big(\{h_j^{(l-1)} : j \in \tilde{\mathcal{N}}(i)\}\big)\Big)\Big) \tag{6}$$

where $\tilde{\mathcal{N}}(i) \subseteq \mathcal{N}(i)$ is a fixed-size sampled neighbourhood and AGG$(\cdot)$ is an aggregation function with mean aggregation being used as given in (7):

$$\text{AGG}\big(\{h_j^{(l-1)}\}\big) = \frac{1}{|\tilde{\mathcal{N}}(i)|} \sum_{j \in \tilde{\mathcal{N}}(i)} h_j^{(l-1)} \tag{7}$$

The concatenation of $h_i^{(l-1)}$ with the aggregated neighbourhood preserves the node's own identity while incorporating neighbour context. Three such layers were stacked, yielding $h_i^{(3)} \in \mathbb{R}^3$.

3) Graph Attention Network (GAT): GAT [3] introduces learnable attention coefficients to weight neighbour contributions. The attention coefficient between node n_i and neighbour n_j at layer l is computed as shown in (8):

$$e_{ij}^{(l)} = \text{LeakyReLU}\Big(\mathbf{a}^\top \Big[W^{(l)} h_i^{(l-1)} \parallel W^{(l)} h_j^{(l-1)}\Big]\Big) \tag{8}$$

where $\mathbf{a} \in \mathbb{R}^{2d_l}$ is a learnable attention vector, $\parallel \cdot \parallel$ denotes concatenation, and $W^{(l)}$ is a shared weight matrix. The raw scores are normalised over the neighbourhood using softmax (shown in (9)):

$$\alpha_{ij}^{(l)} = \frac{\exp\Big(e_{ij}^{(l)}\Big)}{\sum_{k \in \mathcal{N}(i)} \exp\Big(e_{ik}^{(l)}\Big)} \tag{9}$$

Multi-head attention with K independent heads is applied to stabilise training. The node representation is updated as shown in (10):

$$h_i^{(l)} = \sigma\left(\frac{1}{K} \sum_{k=1}^{K} \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(l,k)}\, W^{(l,k)} h_j^{(l-1)}\right) \tag{10}$$

In the implementation carried out during the study, $K = 4$ attention heads were used with two convolutional layers, yielding $h_i^{(2)} \in \mathbb{R}^3$.

4) Classification and Loss: For all three models, the final node representation $h_i^{(L)} \in \mathbb{R}^3$ is passed through a softmax layer to obtain a class probability distribution (as given in (11)):

$$\hat{y}_i = \text{softmax}(h_i^{(L)}) \in \mathbb{R}^3 \tag{11}$$

Training minimises the weighted cross-entropy loss over the labelled node set $\mathcal{V}_L$ (20% of all nodes) as shown in (12):

$$\mathcal{L} = -\sum_{i\in\mathcal{V}_L}\sum_{c\in\mathcal{C}} w_c\, y_{ic}\, \log \hat{y}_{ic} \tag{12}$$

where $y_{ic} \in \{0, 1\}$ is the ground-truth indicator for class c, $\hat{y}_{ic}$ is the predicted probability for class c, and w_c is the class weight defined in (13):

$$w_c = \frac{(\text{count}_c)^{-1.5}}{\sum_{k\in\mathcal{C}}(\text{count}_k)^{-1.5}} \tag{13}$$

The power factor of 1.5 applies stronger upweighting to minority classes (*Bad* and *Nuanced*) than simple inverse frequency ($p = 1.0$) would. Loss is back-propagated using the Adam optimiser with a learning rate scheduler that decays the rate when validation loss plateaus.

E. Experimental Setup

1) Training Regime: The GNNs were trained under a reduced-supervision regime. All 23,873 nodes were labelled, but only 20% (4,774 nodes) were exposed to loss computation during training via the loss function in (12); the remaining 80% participated in message passing but did not contribute to the loss. This is *not* canonical semi-supervised learning, which presupposes genuinely unlabelled data; rather, it is reduced-supervision training that simulates annotation-budget constraints, consistent with the setting studied by Lee et al. [6]. The 4,774 labelled nodes were split 80/10/10 into training (3,819 nodes), validation (477 nodes), and test (478 nodes) subsets.

2) Training Pipeline: A unified training pipeline ensured fair comparison across all three GNNs (Fig. 1). The pipeline iterates over a fixed seed set $S = \{1, 2, 3, 4, 5\}$ to control randomness in weight initialisation, data shuffling, and dropout. GCN and GraphSAGE used three hidden layers each; GAT used two. Training parameters are given in Table V.

TABLE V
TRAINING PARAMETERS

Parameter	GCN	GraphSAGE	GAT
In Channels	768	768	768
Hidden Channels	128	128	128
Out Channels	3	3	3
Attention Heads	3	3	4

Each epoch computes predictions via (11), calculates weighted cross-entropy loss via (12), and updates parameters via backpropagation. Training stops when both validation accuracy and per-epoch loss plateau. The full procedure is given in Algorithm 3.

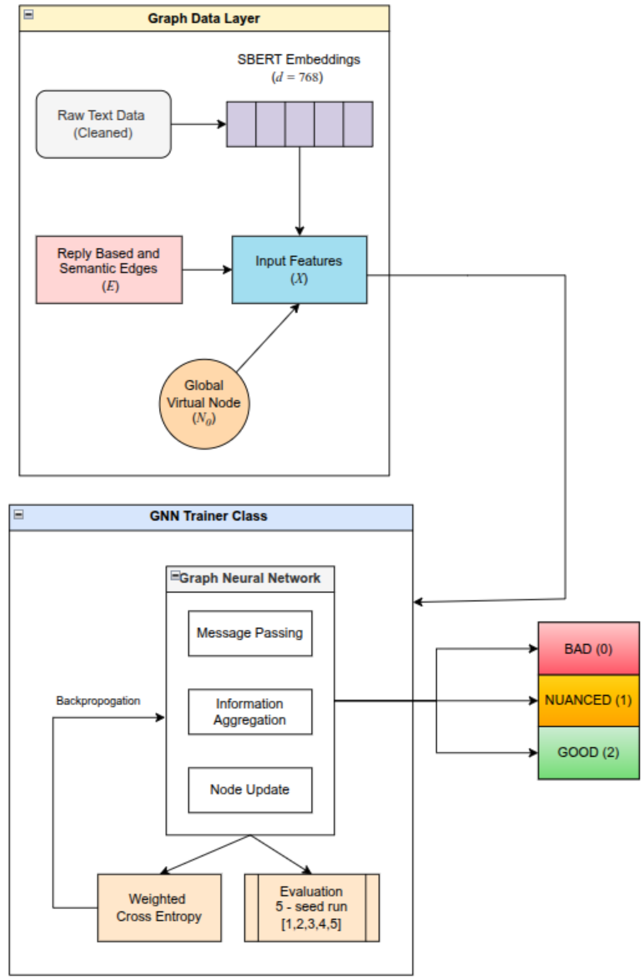


Fig. 1. Conceptual view of the unified training pipeline.

IV. RESULTS AND DISCUSSION

A. Results

Each model trained for up to 1,500 epochs per seed. Metrics were averaged across all seeds; confusion matrices and ROC-AUC curves for the best-performing seed of each model are shown in Figs. 2 and 3. Averaged results appear in Table VI.

TABLE VI
PERFORMANCE AVERAGED ACROSS ALL SEEDS

Metric	GCN	GraphSAGE	GAT
Accuracy	0.578 ± 0.008	0.671 ± 0.002	0.585 ± 0.006
Precision	0.595 ± 0.001	0.647 ± 0.002	0.605 ± 0.003
Recall	0.633 ± 0.001	0.646 ± 0.008	0.633 ± 0.005
F1 Score	0.579 ± 0.006	0.646 ± 0.003	0.549 ± 0.005

The performance values in Table VI reflect the inherent difficulty of the task rather than under-training. All models

Algorithm 3 GNN Training and Evaluation Pipeline

Require: Graph $G(V,E)$, features X, labels Y, seeds S, power $p = 1.5$

Ensure: Mean metrics across seeds; confusion matrix for best seed

Compute w_c per (13); normalise $\sum w_c = 1$
for each seed $r \in S$ **do**
  Set random seed r
  Initialise model M, Adam optimiser, LR scheduler
  for epoch $= 1$ **to** $E_{\max}$ **do**
    $\hat{Y} \leftarrow M(G,X)$ {Forward pass per Eq. ((5))/((6))/((10))}
    Compute $\mathcal{L}$ per (12))
    Update M via backpropagation; update LR
    if validation accuracy and loss have plateaued **then**
      break
    end if
  end for
  Record Accuracy, Precision, Recall, F1, ROC-AUC for seed r
end for
Report mean $\pm$ std; generate confusion matrix for best seed

were trained for up to 1,500 epochs with early stopping, and training loss converged in all runs. The primary factors limiting accuracy are: (1) the subjective and context-dependent nature of psychological impact labels, which introduces irreducible label noise; (2) class imbalance skewed toward the *Good* category; and (3) the restricted size of the training mask (20% of nodes). These are dataset and task characteristics, not modelling failures. GraphSAGE at 67.1% represents a 34-percentage-point improvement over the random baseline (33%) and a 12-percentage-point improvement over the best text-only baseline (55%).

GraphSAGE achieved the highest accuracy (67.1%) and F1-score (0.646), showing the highest true positive rate across all three classes in Fig. 2. This is consistent with its inductive neighbourhood sampling mechanism [2]: by aggregating local features rather than relying on fixed spectral filters (as in GCN, (5)), GraphSAGE is better suited to the heterogeneous degree distribution of a conversational graph where popular posts attract orders of magnitude more replies than average nodes.

Per-class performance for the best GraphSAGE seed (Seed 1) is given in Table VII.

TABLE VII
PER-CLASS PERFORMANCE FOR GRAPHSAGE (BEST SEED)

Class	Precision	Recall	F1-score
Good	0.72	0.75	0.74
Nuanced	0.67	0.61	0.64
Bad	0.57	0.56	0.56

The *Bad* class showed the lowest recall across all three models. This reflects both the class imbalance in social media data (where harmful content is a minority) and the inherent difficulty of distinguishing *Bad* from *Nuanced* content that employs measured but ultimately misleading framing.

Logistic Regression (one-vs-rest) and XGBoost trained on node text features alone each achieved 55% accuracy. The 12–16 percentage point improvement of GraphSAGE over these baselines is attributable to the relational structural information provided by the graph—information that a flat text classifier cannot access regardless of embedding quality.

B. Discussion

1) Impact vs. Sentiment: Sentiment is the degree to which content satisfies the agenda a reader brings to it before consumption—binary and reader-dependent. Psychological impact is the effect the content leaves on the reader independent of prior expectations. The same text can be positive in sentiment while simultaneously reinforcing a harmful worldview. This distinction is not merely definitional: a model trained on sentiment labels will not generalise to impact prediction because the two labels frequently disagree on the same node.

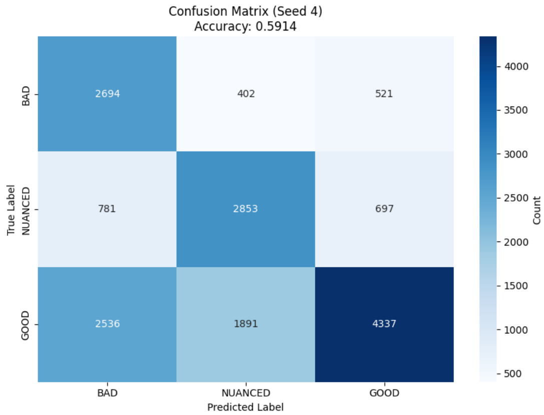
2) Graph Construction: The progression from reply-only edges (E_1, (2)) to the full graph (adding E_2 per (3), then N_0 per (4)) produced a clear step-wise improvement. The reply-only graph was too sparse for meaningful message passing. Semantic edges introduced cross-thread connectivity; the global virtual node then reduced the diameter of remaining disconnected components. Each structural addition was motivated by a specific deficiency in the preceding configuration.

3) Why GraphSAGE Outperforms GCN and GAT: GCN ((5)) assigns equal normalised weight to all neighbours and requires the full graph at training time. GAT (Equations (8)–(10)) assigns learned attention weights but still aggregates over the complete neighbourhood. GraphSAGE (Equations (6)–(7)) samples a fixed-size neighbourhood at each layer, making it robust to the high-degree variance characteristic of social media graphs where thread popularity follows a power law. In the graph constructed during implementation, a handful of popular post nodes have hundreds of reply children while the majority have one or two; GraphSAGE's sampling caps the disproportionate influence of high-degree nodes.

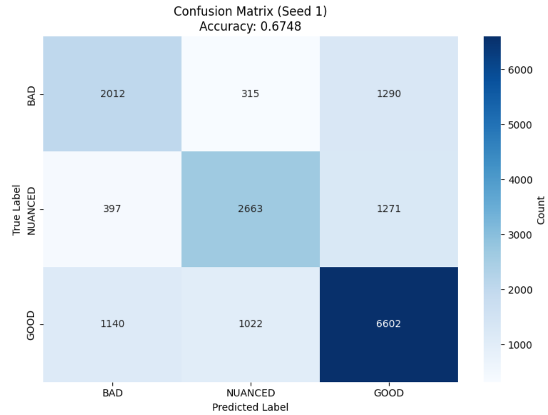
C. Strengths and Limitations

1) Strengths: Conversation-based edges and semantic edges give the GNN access to both local conversational context and the broader topical landscape of the dataset. This inductive bias is unavailable to any node-independent text classifier. The reduced-supervision regime is practically motivated: in real content moderation pipelines, annotation is expensive.

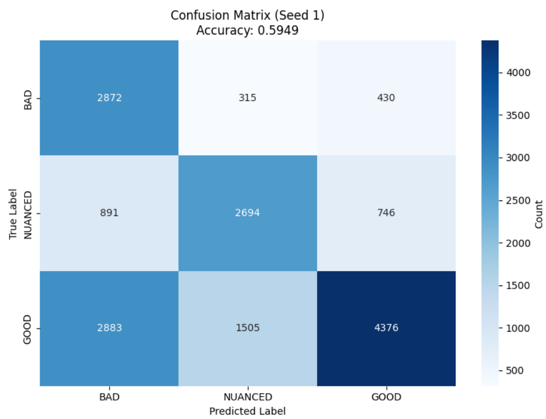
2) Limitations: The $\mathcal{O}(|N|^2)$ pairwise comparison for semantic edge construction limits scalability. The threshold τ must be set before training and may not transfer to corpora with different linguistic characteristics. The dataset is skewed toward the *Good* class; despite inverse-frequency weighting, this affects minority-class recall. The global virtual node may suppress local structural signals. Labelling via zero-shot



(a) GCN



(b) GraphSAGE



(c) GAT

Fig. 2. Confusion matrices representing peak performance across all experimental seed initialisations.

BERT introduces model-dependent bias, partially mitigated by manual review.

3) Reproducibility: The code, data extraction scripts, and processed dataset will be made publicly available upon acceptance.

V. Conclusion

This study proposed a graph-based framework for classifying social media discourse nodes by the psychological impact they leave on the reader. Conversational data from Hacker News and YouTube were structured as a directed heterogeneous graph with reply-based and semantic similarity edges. A global virtual node reduced graph sparsity and enabled graph-wide context propagation. Three GNNs — GCN, GraphSAGE, and GAT — were trained under a reduced-supervision regime. GraphSAGE achieved the strongest performance across all metrics, with accuracy and F1-score consistently higher than GCN and GAT. The improvement over text-only baselines demonstrates that relational structure in the conversational graph carries information no flat text classifier can recover.

Future work will explore jointly modelling content and author behavioural dynamics, scaling semantic edge construction with approximate nearest-neighbour methods, and applying canonical semi-supervised learning with genuinely unlabelled data.

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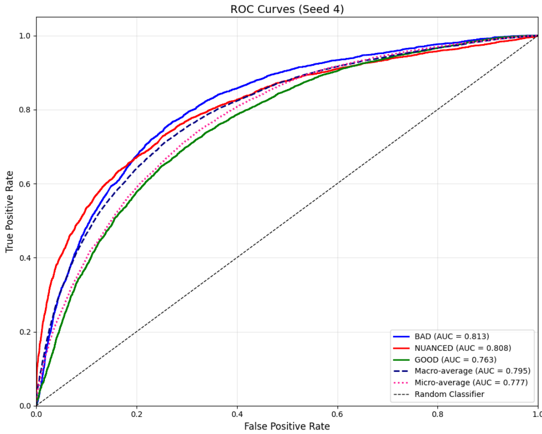
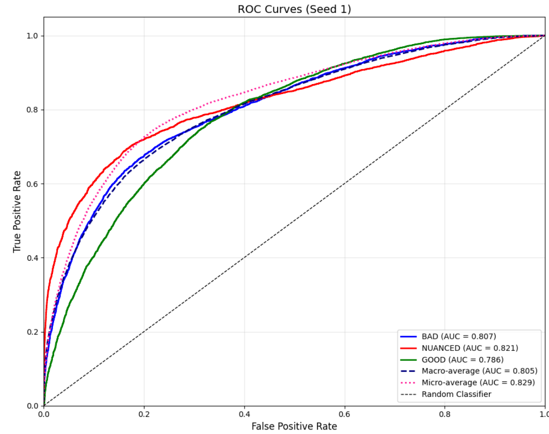
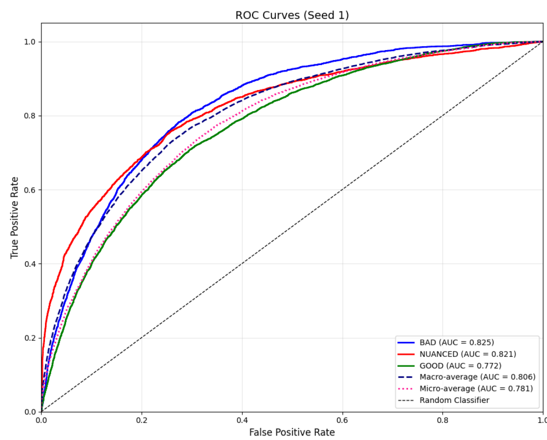


(a) GCN



(b) GraphSAGE



(c) GAT

Fig. 3. ROC-AUC curves representing peak performance across all experimental seed initialisations.

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